

```
!wget -q https://go.dev/dl/go1.21.5.linux-amd64.tar.gz
!tar -C /usr/local -xzf go1.21.5.linux-amd64.tar.gz
import os
os.environ['PATH'] += ':/usr/local/go/bin'
!go version
```

Show hidden output

```
!git clone https://github.com/INMAP-PAVITRA/InMAP-PAVITRA.git
%cd InMAP-PAVITRA
```

Show hidden output

```
!GO111MODULE=on go build ./cmd/inmap
```

Show hidden output

```
!wget -q --show-progress "https://zenodo.org/records/3403934/files/evaldata_v1.6.1.zip?download=1" -O /
```

Show hidden output

```
!unzip -q /content/evaldata_v1.6.1.zip -d /content/
!ls -la /content/evaldata_v1.6.1
```

Show hidden output

```
from google.colab import files
uploaded = files.upload()
```

Show hidden output

```
!pip install netCDF4 -q --break-system-packages 2>/dev/null
import netCDF4 as nc
ds = nc.Dataset('/content/evaldata_v1.6.1/la_test/inmapData.ncf')
print(ds)
print()
for v in ds.variables:
    print(v, ds.variables[v].shape)
```

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```
config_content = '''InMAPData = "/content/evaldata_v1.6.1/la_test/inmapData.ncf"
VariableGridData = "/content/evaldata_v1.6.1/la_test/inmapVarGrid.gob"

EmissionsShapefiles = [
    "/content/evaldata_v1.6.1/la_test/testEmis.shp"
]

EmissionUnits = "tons/year"
HTTPAddress = ":8080"

OutputFile = "/content/InMAP-PAVITRA/laTest/laTest_output.shp"
LogFile = "/content/InMAP-PAVITRA/laTest/logfile.log"

[OutputVariables]
TotalPM25 = "PrimaryPM25 + pNH4 + pSO4 + pNO3 + SOA"
TotalPopD = "(exp(log(1.078)/10 * TotalPM25) - 1) * TotalPop * AllCause / 100000"
```

```

TotalPop = "TotalPop"
PrimPM25 = "PrimaryPM25"
PSO4 = "pSO4"
PNO3 = "pNO3"
PNH4 = "pNH4"
SOA = "SOA"
SOx = "SOx"
NH3 = "NH3"
NOx = "NOx"
BasePM25 = "BaselineTotalPM25"
WindSpeed = "WindSpeed"

[VarGrid]
VariableGridXo = -2004000.0
VariableGridYo = -540000.0
VariableGridDx = 12000.0
VariableGridDy = 12000.0
Xnests = [24]
Ynests = [20]
HiResLayers = 1
GridProj = "+proj=lcc +lat_1=33.000000 +lat_2=45.000000 +lat_0=40.000000 +lon_0=-97.000000 +x_0=0 +y_0=0"
PopDensityThreshold = 0.0055
PopThreshold = 40000.0
PopConcThreshold = 0.000000001

CensusFile = "/content/evaldata_v1.6.1/la_test/census2012.shp"
CensusPopColumns = ["TotalPop"]
PopGridColumn = "TotalPop"

MortalityRateFile = "/content/evaldata_v1.6.1/la_test/MortalityRate.shp"

[VarGrid.MortalityRateColumns]
AllCause = "TotalPop"
'''

with open('/content/laTestConfig.toml', 'w') as f:
    f.write(config_content)

print("Config updated")

```

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```

!mkdir -p /content/InMAP-PAVITRA/laTest
%cd /content/InMAP-PAVITRA
!./inmap run steady --config=/content/laTestConfig.toml

```

[Show hidden output](#)

```

!ls -la /content/InMAP-PAVITRA/laTest/

```

[Show hidden output](#)

```

!pip install geopandas -q --break-system-packages 2>/dev/null
import geopandas as gpd

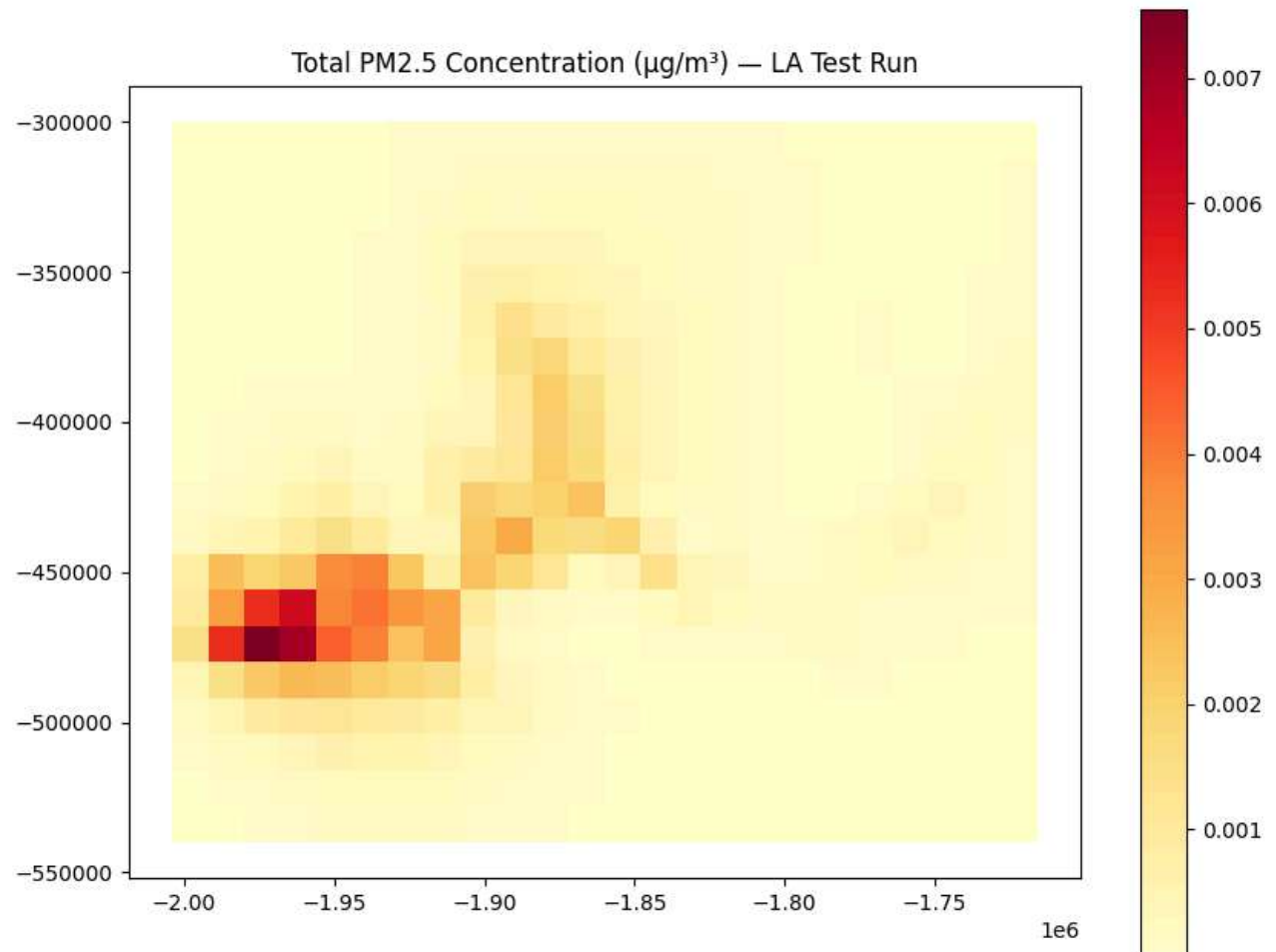
gdf = gpd.read_file('/content/InMAP-PAVITRA/laTest/laTest_output.shp')
print("Columns:", gdf.columns.tolist())
print()
print("Number of grid cells:", len(gdf))
print()
print(gdf[['TotalPM25', 'TotalPop', 'TotalPopD']].describe())

```

[Show hidden output](#)

```
import matplotlib.pyplot as plt

fig, ax = plt.subplots(figsize=(10, 8))
gdf.plot(column='TotalPM25', cmap='YlOrRd', legend=True, ax=ax)
ax.set_title('Total PM2.5 Concentration (µg/m³) — LA Test Run')
plt.show()
```



```
# CHECK STATUS
!cat /content/inmap_run.log
!ps aux | grep inmap
```

```
cat: /content/inmap_run.log: No such file or directory
root      9844  0.0  0.0   7376  3456 ?        S   11:29   0:00 /bin/bash -c ps aux | grep inmap
root      9846  0.0  0.0   6484  2452 ?        S   11:29   0:00 grep inmap
```

```
!tail -20 /content/inmap_run.log
!ps aux | grep inmap
!free -h
```

```
2026/06/16 09:30:41 Parsing output variable expressions...
2026/06/16 09:30:41 Loading emissions shapefile: /content/evaldata_v1.6.1/2005_emissions/bioFireEmis.shp
2026/06/16 09:30:58 Loading emissions shapefile: /content/evaldata_v1.6.1/2005_emissions/Elevated.shp.
root      28615  0.0  0.0   7376  3384 ?        S   09:55   0:00 /bin/bash -c ps aux | grep inmap
root      28617  0.0  0.0   6484  2424 ?        S   09:55   0:00 grep inmap
      total          used          free      shared buff/cache       available
```